

Problem

Design a problem to help other students better understand the design of active high-pass filters when specifying a high-frequency gain and a corner frequency.

Step-by-step solution

Step 1 of 1

Design a problem to help other students better understand the design of active highpass filters when specifying a high-frequency gain and a corner frequency.

Although there are many ways to solve this problem, this is an example based on the same kind of problem asked in the third edition.

Problem

Design an active highpass filter with a high-frequency gain of 5 and a corner frequency of 200 Hz.

Solution

$$\text{High frequency gain} = 5 = \frac{R_f}{R_i} \longrightarrow R_f = 5R_i$$

$$\text{Corner frequency} = \omega_c = \frac{1}{R_i C_i} = 2\pi(200) \text{ rad/s}$$

If we select $R_i = 20 \text{ k}\Omega$, then $R_f = 100 \text{ k}\Omega$ and

$$C = \frac{1}{(2\pi)(200)(20 \times 10^3)} = 39.8 \text{ nF}$$

Therefore, if $R_i = \underline{20 \text{ k}\Omega}$, then $R_f = \underline{100 \text{ k}\Omega}$ and $C = \underline{39.8 \text{ nF}}$